



ZXA10 C300/C350/C320

Optical Access Convergence Equipment

Initial Configuration

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About This Manual

Purpose

This manual provides the initial configuration guide for the ZXA10 C300/C350/C320.

Intended Audience

This manual is intended for:

- Network planning engineers
- Installation and debugging engineers
- On-site maintenance engineers
- System maintenance engineers

What Is in This Manual

This manual contains the following chapters.

Chapter 1, Configuration Flow	Describes the initial configuration flow.
Chapter 2, Configuration Preparations	Describes the software and hardware requirements and how to enter configuration mode of the ZXA10 C300/C350/C320.
Chapter 3, Basic Data Configuration	Describes how to configure in-band Network Management System, out-band Network Management System, physical configurations, and the system time.
Chapter 4, Interconnection Parameters Configuration	Describes how to configure interconnection parameters.
Chapter 5, Basic Services Configuration	Describes how to configure the basic service.

Conventions

This manual uses the following conventions.

	Note: provides additional information about a topic.
---	--

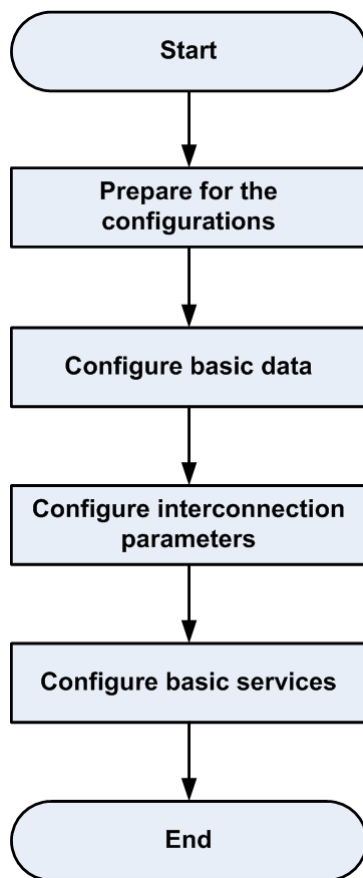
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Chapter 1

Configuration Flow

Figure 1-1 shows the initial configuration flow.

Figure 1-1 Initial Configuration Flow



1. Prepare for the configuration.

The preparations that must be made before the configuration include the operations for meeting software and hardware requirements and entering configuration mode.

2. Configure basic data.

Basic data includes the inband and out-of-band Network Management (NM) configurations, physical configuration, system time, and operation users, which is the basis for configuring the ZXA10 C300/C350/C320 services.

3. Configure interconnection data.

The interconnection data is required for communication with the upper-layer network devices.

4. Configure basic services.

Basic services include the VLAN, xPON service, and NTP service.

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Chapter 2

Configuration Preparations

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Software and Hardware Requirements	2-1
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2.1 Software and Hardware Requirements

Hardware Requirements

The following hardware items are completed:

- Racks, shelves, and cards are installed.
- Cards and shelves are connected properly.
- The power supply is available, and the device is already started.

Software Requirements

The following software item is completed:

The **NM** software is installed.

2.2 Entering Configuration Mode

This procedure describes how to enter configuration mode of the ZX A10 C300/C350/C320. In configuration mode, you can implement initial configurations.

Steps

1. Connect the **CLI** port on the switching and control card to the serial port on the debugging terminal (PC) using a debugging cable.
2. On the debugging terminal, start the HyperTerminal, and set the COM port parameters to their default values.
3. At the **ZXAN>** prompt, run the **enable** command and type the password to enter administrator mode (**ZXAN#**). The initial password is **zxrl0**.
4. In **ZXAN#** mode, run the **configure terminal** to enter global configuration mode (**ZXAN(config)#**).
5. In **ZXAN(config)#** mode, run the following commands to enter the specified configuration mode.

Option	Description
Enter the VLAN interface mode (ZXAN(config-if-vlanid)#)	Run the interface vlan command
Enter the GPON configuration mode (ZXAN(config-gpon)#)	Run the gpon command
Enter the EPON configuration mode (ZXAN(config-epon)#)	Run the epon command
Enter the PON configuration mode (ZXAN(config-pon)#)	Run the pon command
Enter the GPON OLT interface configuration mode (ZXAN(config-if)#)	Run the interface gpon-olt command
Enter the EPON OLT interface configuration mode (ZXAN(config-if)#)	Run the interface epon-olt command
Enter the GPON ONU interface configuration mode (ZXAN(config-if)#)	Run the interface gpon-onu command
Enter the EPON ONU interface configuration mode (ZXAN(config-if)#)	Run the interface epon-onu command
Enter the GPON ONU remote management configuration mode (ZXAN(gpon-onu-mng 1/x/x:x)#)	Run the pon-onu-mng gpon-onu command
Enter the EPON ONU remote management configuration mode (ZXAN(epon-onu-mng)#)	Run the pon-onu-mng epon-onu command

– End of Steps –

Chapter 3

Basic Data Configuration

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Configuring Inband NM Services	3-1
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3.1 Configuring Inband NM Services

In inband **NM** mode, NM interaction information is transmitted through the service channels of the device, which brings a flexible network architecture and saves costs because of no additional devices. However, it is not convenient for maintenance.

Prerequisite

The uplink cards are added.

Steps

1. In **ZXAN#** mode, run the **configure terminal** command to enter global configuration mode.
2. In **ZXAN(config)#** mode, run the **vlan** command to create the inband NM **VLAN**.
3. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink port mode. Run the **switchport vlan** command to add the uplink port to the network management VLAN.
4. In **ZXAN(config)#** mode, run the **interface vlan** command to enter VLAN interface mode. Run the **ip address** command to configure the inband NM IP address.
5. In **ZXAN(config)#** mode, run the **ip route** command to configure inband **NM** routing information.
6. In **ZXAN(config)#** mode, run the **snmp-server community** command to configure the **SNMP** community name.

**Note:**

The SNMP community name must be the same as that configured on the Network Management Server (NMS).

7. In **ZXAN(config)#** mode, run the **snmp-server host** command to configure the IP address of the NMS.
8. In **ZXAN#** mode, run the **write** command to save the configuration data.

– End of Steps –

Example

For the inband NM configurations, refer to [Table 3-1](#).

Table 3-1 Inband NM Configurations

Parameter	Data
NE type	ZXA10 C300
Uplink port	gei_1/20/1
Inband NE VLAN	VLAN ID:1000
Inband NM IP address	10.1.1.1/24
Next-hop IP address	10.1.1.254/24
NMS (SNMP server)	IP address: 10.2.1.1/24 Version: 2c Community name: public (default) Alarm level: NOTIFICATIONS target-addr-name: zte

```

ZXAN#configure terminal
%Info 20272: Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#vlan 1000
ZXAN(config-vlan)#exit
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 1000 tag
ZXAN(config-if)#exit
ZXAN(config)#interface vlan 1000
ZXAN(config-if-vlan1000)#ip address 10.1.1.1 255.255.255.0
ZXAN(config-if-vlan1000)#exit
ZXAN(config)#ip route 10.2.1.1 255.255.255.255 10.1.1.254
ZXAN(config)#snmp-server community public view AllView rw
ZXAN(config)#snmp-server host 10.2.1.1 version 2c public enable NOTIFICATIONS
target-addr-name zte

```

```
ZXAN(config)#exit  
ZXAN#write
```

**Note:**

All configurations in this section take the ZXA10 C300 as an example.

- For the ZXA10 C350 the slot number of PON/P2P interfaces is 2 - 5 and 8 - 10, and the slot number of Ethernet uplink interfaces is 0 and 1.
- For the ZXA10 C320, the slot number of PON/P2P interfaces is 1 and 2, and the slot number of Ethernet uplink interfaces is 3 and 4.

3.2 Configuring Out-of-Band NM Services

In out-of-band NM mode, administrative information is transmitted through the non-service channel, so that the management channel is separated from the service channel. Compared with inband NM mode, this mode provides more reliable channels for device management. If the ZXA10 C300 is faulty, the [NMS](#) can locate it information in time and monitor it in real time.

Steps

1. In **ZXAN#** mode, run the **configure terminal** command to enter global configuration mode.
2. In **ZXAN(config)#** mode, run the **interface mng1** command to enter management interface mode.
3. Run the **ip address** command to set the out-of-band management IP address.
4. In **ZXAN(config)#** mode, run the **ip route** command to configure out-of-band [NM](#) routing information.
5. In **ZXAN(config)#** mode, run the **snmp-server community** command to configure the [SNMP](#) community name.

**Note:**

The SNMP community name must be the same as that configured on the Network Management Server ([NMS](#)).

6. In **ZXAN(config)#** mode, run the **snmp-server host** command to configure the IP address of the NMS.

7. In **ZXAN#** mode, run the **write** command to save the configuration data.

– End of Steps –

Example

For the out-of-band NM configurations, refer to [Table 3-2](#).

Table 3-2 Out-of-Band NM Configurations

Parameter	Data
NE type	ZXA10 C300
Out-of-band NM IP address	11.1.1.1/24
Next-hop IP address	11.1.1.254/24
NMS (SNMP server)	IP address: 11.2.1.1/24 Version: 2c Community name: public (default) Alarm level: NOTIFICATIONS target-addr-name: zte

```
ZXAN#configure terminal
%Info 20272: Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#interface mng1
ZXAN(config-if)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if)#exit
ZXAN(config)#ip route 11.2.1.0 255.255.255.0 11.1.1.254
ZXAN(config)#snmp-server community public view AllView rw
ZXAN(config)#snmp-server host 11.2.1.1 version 2c public enable NOTIFICATIONS
target-addr-name zte
ZXAN(config)#exit
ZXAN#write
```

3.3 Physical Configurations

Physical configurations include the configuration about the following:

- Rack
- Shelf
- Card
- PnP function
- Fan
- Switch over function

Context

For the rack types and shelf types supported by the ZXA10 C300/C350/C320, refer to [Table 3-3](#).

Table 3-3 ZXA10 C300/C350/C320 Rack and Shelf Types

NE Type	Rack Type	Shelf Type
ZXA10 C300	ETSI 21, IEC 19	ETSI_SHELF, IEC_SHELF
ZXA10 C350	C350MBRack	C350MB_SHELF
ZXA10 C320	C320rack	C320_shelf

For the card configurations for a 21-inch ZXA10 C300 shelf, refer to [Table 3-4](#).

Table 3-4 Card Configurations for a 21-Inch ZXA10 C300 Shelf

Slot ID	Card Type
0/1	Power card
2–9, 12–19	PON interface card, TDM interface card, and Ethernet interface card
10/11	Switching and control card
20	Common interface card
21/22	Uplink card

For the card configurations for a 19-inch ZXA10 C300 shelf, refer to [Table 3-5](#).

Table 3-5 Card Configurations for a 19-Inch ZXA10 C300 Shelf

Slot ID	Card Type
0/1	Power card
2–9, 12–17	PON interface card, TDM interface card, and Ethernet interface card
10/11	Switching and control card
18	Common interface card
19/20	Uplink card

For The card configurations for a shelf, refer to [Table 3-6](#).

Table 3-6 Card Configurations for a ZXA10 C350 Shelf

Slot ID	card Type
0/1	Uplink card
2–5, 8–9	PON interface card and P2P interface card
6/7	Switching and control card
10	PON interface card, P2P interface card, and common interface card
11/12	Uplink card

For the card configuration about a ZXA10 C320 shelf, refer to [Table 3-7](#).

Table 3-7 Card Configurations for a ZXA10 C320 Shelf

Slot ID	Card Type
1–2	PON interface card, and P2P interface card
3/4	Switching and control card

For the card status, refer to [Table 3-8](#).

Table 3-8 Card Status Description

Status	Description
INSERVICE	The card is operating properly.
CONFIGING	The card is being configured.
CONFIGFAILED	The service configuration for the card fails.
DISABLE	The card is added logically and is installed in the shelf, but the system fails to receive information about the card.
HWONLINE	The card is installed in the shelf, but its version is incorrect, so that it does not operate properly.
OFFLINE	The card is added logically, but it is installed in the shelf physically.
STANDBY	The card is in standby status.
TYPEMISMATCH	The card type is different from the configured type.
NOPOWER	The power card is not powered on.

Steps

1. In **ZXAN#** mode, run the **configure terminal** command to enter global configuration mode.
2. In **ZXAN(config)#** mode, run the **add-rack** command to add the rack.



Note:

The ZXA10 C300/C350/C320 support only one rack, and thus the **rackno** parameter can only be 1.

3. In **ZXAN(config)#** mode, run the **add-shelf** command to add the shelf.

**Note:**

The ZXA10 C300/C350/C320 support only one shelf, and thus the **shelfno** parameter can only be 1.

4. Add cards.

Option	Description
Automatic add cards	In ZXAN(config)# mode, run the set-pnp enable command to enable the PnP function
Manually add cards	In ZXAN(config)# mode, run the add-card command to add a card

5. (Optional) Run the following command to configure the relational task.

Option	Description
Delete a card	In ZXAN(config)# mode, run the del-card command
Reset a card	In ZXAN# mode, run the reset-card command
Switch over the active and standby switching and control cards	In ZXAN# mode, run the swap command

6. Configure the fan parameters.

Option	Description
Configure the operating mode for fans	In ZXAN(config)# mode, run the fan control command
Set the rotational speed percentage for each level	In ZXAN(config)# mode, run the fan speed-percent-set command
Configure the temperature threshold	In ZXAN(config)# mode, run the fan temp-zone command

7. In any mode, query the configuration information.

Option	Description
Query the information about all cards	Run the show card command
Query the information about a specified board	Run the show card slotno command
Query the fan configuration	Run the show fan command

– End of Steps –

Example

Take the ZXA10 C300 19-inch shelf as an example.

- The following example shows how to configure the rack:

```
ZXAN#configure terminal
%Info 20272: Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#
ZXAN(config)#add-rack rackno 1 racktype IEC19
```

- The following example shows how to configure the shelf:

```
ZXAN(config)#add-shelf shelfno 1 shelftype IEC_SHELF
```

- The following example shows how to configure a card:

```
ZXAN(config)#add-card slotno 3 ETGO
ZXAN(config)#add-card slotno 5 GTGH
ZXAN(config)#add-card slotno 19 GUSQ
```

- The following example shows how to enable the PnP function:

```
ZXAN(config)#set-pnp enable
```

- The following example shows how to delete a card:

```
ZXAN(config)#del-card slotno 3
Confirm to delete card? [yes/no]:yes
```

- The following example shows how to reset a card:

```
ZXAN#reset-card slotno 5
Confirm to reset card? [yes/no]:y
```

- The following example shows how to switch over the active and standby switching and control cards:

```
ZXAN#swap
Confirm to master swap? [yes/no]:y
```

- The following example shows how to configure fan parameters:

```
ZXAN(config)#fan control temp_level 30 40 50 60
ZXAN(config)#fan speed-percent-set 30 40 50 75
ZXAN(config)#fan temp-zone high-threshold 65
```

- The following example shows how to query the fan configurations:

```
ZXAN(config)#show fan
Shelf                : 1
epm                  :
FanControlType        : temperature-control
TemperatureThreshold   : 30  40  50  60 (deg c)
FanSpeedLevelPercent   : 30%  40%  50%  75%
CriticalTemperatureThreshold : 160(deg c)
HighTemperatureThreshold : 65(deg c)
LowTemperatureThreshold  : -20(deg c)
Environment Temperature : 43(deg c)
```

```

EnvTemperatureProtect      : disable
EnvPowerMode               : Mains Supply
CardCriticalTempProtection : enable
EnvPowerOffProtect         : disable
Upper Fanboard Status      : online
Fan SN                     :
FAN FW Version             :
Fan Temperature            : N/A. (deg c)
All fan units actual status:

```

```

-----
FanUnitId   SpeedLevel   ActualSpeed(RPM)
-----
1           3            2500
2           3            2500
3           3            2500
-----

```

- The following example shows how to query information about all cards:

```

ZXAN#show card
Rack Shelf Slot CfgType RealType Port HardVer SoftVer      Status
-----
1    1    0    PRWH    PRWH    0    V1.0.0           INSERVICE
1    1    5    GTGH    GTGHG   16   V1.0.0 V2.1.0          INSERVICE
1    1    6    GTGO    GTGOE   8    V1.0.0 V2.1.0          INSERVICE
1    1    10   SCXN    SCXN    4    V1.0.0 V2.1.0          INSERVICE
1    1    11   SCXN    SCXN    4    V1.0.0 V2.1.0          STANDBY
1    1    19   GUSQ    GUSQ    4    V1.0.0 V2.1.0          INSERVICE

```

- The following example shows how to display information about the specified card:

```

ZXAN(config)#show card slotno 10
Config-Type      : SCXN      Status      : INSERVICE
Port-Number      : 4
Cpu-Alarm-Threshold : 100%
Mem-Alarm-Threshold : 100%

Real-Type        : SCXN      Serial-Number :
Phy-Mem-Size     : 2048MB    Main-CPU      : PowerPC Processor
M-CODE           : 120902    Hardware-VER  : V1.0.0
Cpld-VER         : V1.9      Fpga-VER      : V2.1.0
OtherfirmwareVER:

BootROM-VER      : V4.0.3    2014-07-17 14:13:38
Software-VER     : V2.1.0    2016-11-04 12:02:12
Cpu-Usage        : 14%
Mem-Usage        : 32%
C-Code           : FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF

```

```
Uptime           : 2 Days, 0 Hours, 18 Minutes, 3 Seconds
Last restart reason : Power-off
```

3.4 Configuring the System Time

After the system time is configured, you can query the [CLI](#) logs and alarms in specific time for troubleshooting.

Context

The internal software maintains the system time for the ZXA10 C300/C350/C320. When the ZXA10 C300/C350/C320 is powered on, the system acquires the hardware clock and initializes the system time of the ZXA10 C300/C350/C320.

Steps

1. In **ZXAN(config)#** mode, run the **clock timezone** command to configure the time zone.
2. In **ZXAN#** mode, run the **clock set** command to configure the system time.

– End of Steps –

Example

```
ZXAN(config)#clock timezone utc 8
ZXAN(config)#exit
ZXAN#clock set 08:00:00 mar 7 2011
ZXAN#show clock
08:01:55 Mon Mar 7 2011 utc
```

3.5 User Management

Users refer to the personnel who configure and maintain the ZXA10 C300/C350/C320 after logging in to it through the CLI terminals, including the serial port terminals, Telnet, or [SSH](#).

Context

In user management, user permissions are divided into 16 levels (0–15). For details, refer to [Table 3-9](#).

Table 3-9 User Permission Descriptions

User Level	Permission Description
0–1	When a user of level 0 or 1 logs in to the device and enters operator mode, the user can enter the enable command and the password to raise the user level to level 15, and perform any operations.
2–9	When a user of level 2 through 9 logs in to the device and enters administrator mode, the user can use commands of levels 0 through 9.

User Level	Permission Description
10–15	When a user of level 10 through 15 logs in to the device and enters administrator mode, the user can use commands of levels 0 through 15. The user can manage and maintain the device and user accounts.

- A user of level 0 can use the commands of only level 0. A user of level 1 can use the commands of levels 0 and 1. A user of level 2 can use the commands of levels 0 through 2, and so on. A user of level 15 can use the commands of levels 0 through 15.

When you create a user, you must configure the user attributes, including the user account, password, and permissions.

Only the users of levels 10 through 15 can create other users. For user attributes, refer to [Table 3-10](#).

Table 3-10 User Attributes

User Attribute	Description
Account	Username, consisting of a maximum of 1 to 16 printable characters except spaces. The username is case sensitive and must be unique.
Password	Consists of 3 to 16 characters. The password is case sensitive.
Maximum number of sessions	A maximum of 16 sessions are supported. Range: 1–16.
Permissions	User permissions are divided into 16 levels. Range: 0–15.

The ZXA10 C300/C350/C320 supports a maximum of 20 users.

Steps

1. In **ZXAN(config)#** mode, run the **service password-encryption** command to enable the password encryption function.
2. In **ZXAN(config)#** mode, run the **username** command to create a user.
3. In **ZXAN(config)#** mode, run the **username** command to modify user attributes.
4. In **ZXAN(config)#** mode, run the **no username** command to delete a user.
5. In **ZXAN#** mode, run the **clear tcp line** command to disconnect a TCP user.
6. In **ZXAN#** mode, run the **clear tcp connect** command to disconnect a TCP connection.

– End of Steps –

Example

- The following example shows how to create a user.

1. Enable the password encryption function:

```
ZXAN(config)#service password-encryption
```

2. Create a user of level 10. The username and password are `abc`.

```
ZXAN(config)#username abc password abc privilege 10
```

3. (Optional) Display user configurations:

```
ZXAN(config)#show username
cli user global configuration
-----
suspend mode: none

name sessions pri OperStatus login-begin login-end expire-date
status
-----
zte      16     1   Normal    00:00:00   23:59:59   2099-12-31 23:59:59
Enable
abc      16    10   Normal    00:00:00   23:59:59   2099-12-31 23:59:59
Enable
```



Note:

If the password encryption function is enabled before a user is added, the encrypted password is displayed after the **show username** command is executed.

- Modify user attributes.

1. Modify the password of the `abc` user to `123`, and user level to `15`:

```
ZXAN(config)#username abc password 123 privilege 15
```

2. (Optional) Display user configurations:

```
ZXAN(config)#show username
cli user global configuration
-----
suspend mode: none

name sessions pri OperStatus login-begin login-end expire-date
status
-----
zte      16     1   Normal    00:00:00   23:59:59   2099-12-31 23:59:59
Enable
abc      16    15   Normal    00:00:00   23:59:59   2099-12-31 23:59:59
Enable
```

- The following example shows how to delete a user.

1. Delete a user:

```
ZXAN(config)#no username abc
```

2. (Optional) Display user configurations:

```

ZXAN(config)#show username
cli user global configuration
-----
suspend mode: none

name sessions pri OperStatus login-begin login-end expire-date
status
-----
zte          16      1   Normal      00:00:00    23:59:59    2099-12-31 23:59:59
Enable

```

- The following example shows how to disconnect an online user.

1. Display online users:

```

ZXAN#show users

```

Line	User	Privilege	Host(s)	Idle	Location
* 66 vty 0	zte	15	idle	00:00:00	10.60.197.21

2. Display the current TCP connections:

```

ZXAN#show tcp brief
ZXAN(config)#show tcp brief

```

TCB	Local Address	Foreign Address	State
1731109028	10.63.192.251.23	10.60.197.21.2375	ESTAB

3. Run the following commands to disconnect an online user:

- Disconnect a user:

```

ZXAN#clear tcp line 69

```

- Disconnect the TCP connection:

```

ZXAN#clear tcp connect 10.1.1.1 23 10.2.1.1 3641

```

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Chapter 4

Interconnection Parameters Configuration

This procedure describes how to configure interconnection parameters to ensure proper communication between the ZX A10 C300/C350/C320 and the upper-layer network devices.

Steps

1. In global configuration mode, run the **vlan** command to create the inband **NM VLAN**.
2. In interface mode of the uplink port, configure the speed, physical operating mode, and port mode (Hybrid, Trunk, or Access).
3. In interface mode of the uplink port, run the **switchport vlan** command to configure the management VLAN.
4. In VLAN interface mode, run the **ip address** command to configure the management IP address.
5. In global configuration mode, run the **ip route** command to configure route information for the inband NM IP address.
6. In administrator mode, run the **ping x.x.x.x** command to ping the IP address to the gateway or the upper-layer network device.

If the Ping is successful, the communication between the ZX A10 C300/C350/C320 and the upper-layer network device is proper. Otherwise, check the uplink network data and the physical connection between them.

– End of Steps –

Example

Table 4-1 lists an example of connection configuration data (takes the ZX A10 C300 as an example).

Table 4-1 Connection Configuration Data

Item	Data
NE type	ZXA10 C300
VLAN ID	1000
Uplink interface	gei_1/20/1
Uplink port mode	trunk

Item	Data
In-band NM IP address	10.1.1.1/24
Default gateway	10.1.1.254/24
Peer end IP address	10.2.1.1/24

The configuration is shown as below:

```
ZXAN(config)#vlan 1000
ZXAN(config-vlan1000)#exit
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport mode trunk
ZXAN(config-if)#switchport vlan 1000 tag
ZXAN(config-if)#exit
ZXAN(config)#interface vlan 1000
ZXAN(config-if-vlan1000)#ip address 10.1.1.1 255.255.255.0
ZXAN(config-if-vlan1000)#exit
ZXAN(config)#ip route 10.2.1.0 255.255.255.0 10.1.1.254
ZXAN(config)#exit
ZXAN#ping 10.2.1.1
sending 5,100-byte ICMP echos to 10.2.1.1,
timeout is 2 seconds.!!!!
Success rate is 100 percent (5/5),round-trip
min/avg/max= 0/0/0 ms.
```

Chapter 5

Basic Services Configuration

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5.1 VLAN Configuration

The Virtual Local Area Network ([VLAN](#)) is a technology that implements virtual workgroups by dividing the physical equipment in a LAN into several logical network segments. The IEEE issued the IEEE 802.1q standard in 1999 to normalize the VLAN solution.

The ZX10 C300/C350/C320 supports a maximum of 4094 VLANs.

For VLAN applications, refer to [Table 5-1](#).

Table 5-1 VLAN Applications

VLAN Application	Description
Basic VLAN	Used to isolate ports.
Service port VLAN	Used to implement VLAN translation at the ONU level.
TLS VLAN	Used to add an SVLAN to a packet to implement the Transparent LAN Service (TLS) whatever the user access mode is, or no matter whether the upstream packet has a VLAN tag, or whatever the VLAN tag is.
VLAN cross connection	Used to set the special channel for the user port and uplink port. The packets are forwarded in 1:1 mode in accordance with the VLAN ID.
Flexible QinQ	Used to add different SVLANs based on different data streams to extend the VLAN.
VLAN translation	Implements translation from a customer VLAN to a network VLAN and adds different SVLANs based on user-side VLANs.

**Note:**

All configurations in this section take the ZXA10 C300 as an example. For the ZXA10 C320, the slot number of PON/P2P interfaces is 1 and 2, and the slot number of Ethernet uplink interfaces is 3 and 4.

5.1.1 Configuring an Uplink Port VLAN

By configuring an uplink port **VLAN**, you can logically divide ports into different network segments to control the communication between ports.

Steps

1. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure an uplink port VLAN.

**Note:**

When you configure the uplink port VLAN, the system automatically creates the corresponding VLAN.

– End of Steps –

Example

The following example shows how to configure a port VLAN in uplink interface configuration mode.

```
ZXAN(config)#interface gei_1/19/1
ZXAN(config-if)#switchport vlan 2-100 tag
```

5.1.2 Configuring a Service Port VLAN

By configuring a service port **VLAN** on the PON-ONU interface, you can implement VLAN translation at the **ONU** level.

Context

The service port configuration of the ZXA10 C300/C350/C320 supports the following applications:

- Adding **CVLAN** + SVLAN to untagged packets
- Adding **SVLANs** to user VLANs in accordance with the user VLAN range
- Translating a user VLAN to VLAN + SVLAN
- Translating a user VLAN to VLAN + SVLAN based on the Ethernet protocol type
- Translating a user VLAN to VLAN + SVLAN based on 802.1p priorities
- Translating a user VLAN to VLAN + SVLAN based on combinations (user VLAN, Ethernet protocol type, and 802.1p priority)
- Modifying SVLAN 802.1p priorities
- **TLS** VLAN

Steps

1. In **ZXAN(config)#** mode, run the **interface gpon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **service-port** command to configure a service port VLAN.

– End of Steps –

Example

1. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 7 vlan 8 svlan 9
ZXAN(config-if)#service-port 2 vport 1 other-all tls-vlan 501
```

2. (Optional) The following example shows how to display the configured service port VLANs:

```
ZXAN(config-if)#show service-port interface gpon-onu_1/5/1:1
Interface gpon-onu_1/5/1:1
Sport Vport BeginVid EndVid OuterVid InnerVid UserPrio Etype Vlan Cos SVlan SCo
s Tls TlsSVid Mode PacketFilter Ingress Egress Queue Status Enable
-----
-----
1      1      7      7      --      --      --      --      8      --  9      --
--      --      1:1  --      --      --      --      --      YES
2      1      --      --      --      --      --      --      --      --  --      --
501    --      --      --      --      --      --      --      YES
Sport total number:
2
```

5.1.3 Configuring the VLAN Cross Connection

This procedure describes how to configure the **VLAN** cross connection to implement 1:1 VLAN forwarding.

Context

The VLAN cross connection is a special channel for a user port and an uplink port. After the VLAN cross connection is configured, packets are forwarded in 1:1 mode in accordance with the VLAN ID but not forwarded in MAC + VLAN mode.

For the VLAN cross exchange, data is exchanged in accordance with:

- **SVLAN**
- **CVLAN** + SVLAN dual tags

Steps

1. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure an uplink port VLAN.
3. In **ZXAN(config)#** mode, run the **interface gpon-onu** command to enter ONU interface configuration mode.
4. In **ZXAN(config-if)#** mode, run the **service-port** command to configure a service port VLAN.
5. In **ZXAN(config-vlanid)#** mode, run the **xconnect enable cvlan** command to set VLAN translation to 1:1 VLAN translation.

– End of Steps –

Example

1. The following example shows how to configure a port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/19/1
ZXAN(config-if)#switchport vlan 5 tag
ZXAN(config-if)#exit
```

2. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 3 svlan 5
ZXAN(config-if)#exit
```

3. The following example shows how to set VLAN translation to 1:1 VLAN translation in VLAN configuration mode:

```
ZXAN(config)#vlan 5
ZXAN(config-vlan5)#xconnect enable cvlan 3
```

4. (Optional) The following example shows how to display the VLAN cross connection configuration:

```
ZXAN(config-vlan5)#show vlan-xconnect detail

User-Port                Uplink-Port  Svlan  Cvlan  Status
```

```
-----
gpon-onu_1/5/1:1 vport 1      gei_1/19/1    5      3      --
```

5.1.4 Configuring Flexible QinQ

This procedure describes how to configure the flexible QinQ rule to add different external-layer VLAN tags for different service streams to implement service distribution.

Context

For the flexible QinQ function of the ZXAN C300/C350/C320, the service stream classification is based on the following options:

- CoS priority of a service stream
- User CVLAN range
- Ethernet type (Ether-type) of a service stream
- Combination of CVLAN and CoS
- Combination of CVLAN and a Ethernet type

For the GPON, it is recommended to implement the flexible QinQ function using the ONU interface service-port.

Steps

1. In **ZXAN(config)#** mode, run the **vlan-smart-qinq enable** command to enable the QinQ function.
2. In **ZXAN(config)#** mode, run the **interface epon-olt** command to enter OLT interface configuration mode.
3. In **ZXAN(config)#** mode, run the **vlan-smart-qinq enable** command to enable the QinQ function of the OLT interface.
4. In **ZXAN(config)#** mode, run the **vlan-smart-qinq ingress-port** command to configure the QinQ rule.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure an uplink port VLAN.

– End of Steps –

Example

1. The following example shows how to enable the flexible QinQ function in global configuration mode:

```
ZXAN(config)#vlan-smart-qinq enable
```

2. The following example shows how to enable the flexible QinQ function in OLT interface configuration mode:

```
ZXAN(config)#interface epon-olt_1/6/1
ZXAN(config-if)#vlan-smart-qinq enable
ZXAN(config-if)#exit
```

3. The following example shows how to configure the VLAN Smart QinQ rule:

```
ZXAN(config)#vlan-smart-qinq ingress-port epon-olt_1/6/1 cvlan 200 to 299 svlan 4000
newcos 6
```

4. The following example shows how to configure a port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/19/2
ZXAN(config-if)#switchport vlan 4000 tag
ZXAN(config-if)#exit
```

5. (Optional) The following example shows how to display the QinQ rule configuration:

```
ZXAN(config)#show vlan-smart-qinq
smart-qinq item num(used): 1
Ingress-Port      Begin- End-   EType  Cos Svlan NewCos Status
                  Vid    Vid
-----
epon-olt_1/6/1    200    299    ----   --  4000  6      --
```

5.1.5 Configuring VLAN Translation

The following example shows how to configure a [VLAN](#) translation entry to convert a customer VLAN to a network VLAN to distribute service traffic. If you do not specify N:1 VLAN translation, 1:1 VLAN translation is used by default, which means one customer VLAN to one network VLAN.

Context

For the [GPON](#), it is recommended to implement the VLAN translation function through the [ONU](#) interface service-port.

Steps

1. In **ZXAN(config)#** mode, run the **vlan-translate ingress-port** command to configure a VLAN translation entry.
2. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure an uplink port VLAN.

– End of Steps –

Example

1. The following example shows how to configure a VLAN translation entry:

- Configure 1:1 VLAN translation:

```
ZXAN(config)#vlan-translate ingress-port epon-olt_1/6/1 user-vlan 101 vlan 201 1:1
```

- Configure N:1 VLAN translation:

```
ZXAN(config)#vlan-translate ingress-port epon-olt_1/6/1 user-vlan 102 vlan 202 N:1
ZXAN(config)#vlan-translate ingress-port epon-olt_1/6/1 user-vlan 103 vlan 202 N:1
```

- Configure 1:1 VLAN translation and add an SVLAN:

```
ZXAN(config)#vlan-translate ingress-port epon-olt_1/6/1 user-vlan 100 vlan 200
svlan 300 1:1
```

- Configure N:1 VLAN translation and add an SVLAN:

```
ZXAN(config)#vlan-translate ingress-port epon-olt_1/6/1 user-vlan 105 vlan 205
svlan 305 N:1
ZXAN(config)#vlan-translate ingress-port epon-olt_1/6/1 user-vlan 106 vlan 205
svlan 305 N:1
```

- The following example shows how to configure a port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/19/2
ZXAN(config-if)#switchport vlan 201,202,300,305 tag
ZXAN(config-if)#exit
```

- (Optional) The following example shows how to display a VLAN translation configuration:

```
ZXAN(config)#show vlan-translate
```

Ingress-Port	User-Vlan	OuterVid	InnerVid	Vlan	Svlan	TransMode	Status

epon-olt_1/6/1	101	--	--	201	--	--	--
epon-olt_1/6/1	102	--	--	202	--	--	--
epon-olt_1/6/1	103	--	--	202	--	--	--
epon-olt_1/6/1	100	--	--	200	300	--	--
epon-olt_1/6/1	105	--	--	205	305	--	--
epon-olt_1/6/1	106	--	--	205	305	--	--

5.1.6 Configuring VLAN Translation for Broadcast Traffic

After [VLAN](#) translation is configured on the broadcast [LLID](#) or the broadcast [GEM](#) port, the ZXA10 C300/C350/C320 can implement VLAN processing for the downlink traffic on the broadcast LLID and broadcast GEM port.

Context

The VLAN translation configuration on the Service-port or switchport and the [EPON OLT](#) interface takes effect only for the traffic on the unicast LLID and unicast GEM Port, but not for that on the broadcast LLID and broadcast GEM Port.

- On the EPON interface card, the broadcast LLID transmits downlink multicast, broadcast, and flooding traffic, and the unicast LLID transmits other downlink traffic and all uplink traffic.
- On the GPON interface card, the broadcast GEM Port transmits only downlink multi-cast, and the unicast GEM Port transmits other downlink traffic and all uplink traffic.

Steps

1. In **ZXAN(config)#** mode, run the **mvlan-translate** command to configure a VLAN translation entry.
2. In **ZXAN(config)#** mode, run the following commands to configure the SCB VLAN processing on the OLT interface:
 - a. In **ZXAN(config)#** mode, run the **vlan-scb-action interface** command to remove the VLAN tag of the downlink SCB traffic.
 - b. In **ZXAN(config)#** mode, run the **vlan-scb-action interface** command to remove the VLAN tag of the downlink SCB traffic.

– End of Steps –

Example

1. The following example shows how to configure a global unicast VLAN translation entry:

```
ZXAN(config)#mvlan-translate 4000 to 3006
```
2. Configure the SCB VLAN processing on the EPON OLT interface.
 - a. The following example shows how to remove the VLAN tag of the downlink SCB traffic:

```
ZXAN(config)#vlan-scb-action interface epon-olt_1/6/1 vlan-remove 4094
```
 - b. The following example shows how to configure the VLAN translation entry for the downlink SCB traffic:

```
ZXAN(config)#vlan-scb-action interface epon-olt_1/6/1 vlan-translate svlan 10  
vlan 20 user-vlan 30
```

5.2 EPON/10G-EPON Service Configuration

EPON access is a flexible access technology that provides super bandwidth access in both broadband and narrowband service environment. It supports multiple rate modes and uses a single optical fiber to provide subscribers with the voice, data, and video services.

The principles and configuration of the 10G-EPON services and the EPON services are the same except that the 10G-EPON provides a data bandwidth of 10 Gbps.

5.2.1 Configuring an EPON ONU Type Profile

Before authentication of an EPON ONU, you need to create an ONU type profile if an ONU type does not exist.

Context

The ZXAN10 C300 supports the following default EPON ONU types:

- ZTE-9806
- ZTE-D400, ZTE-D402, ZTE-D420, ZTE-D421, and ZTE-D422
- ZTE-F401, ZTE-F420, ZTE-F425, ZTE-F429, ZTE-F430, and ZTE-F435
- ZTE-F500
- ZTE-F803, ZTE-F820, ZTE-F821, and ZTE-F822

You can use the **show onu-type epon** command to query the default EPON ONU types.

Steps

1. In **ZXAN(config)#** mode, run the **pon** command to enter PON configuration mode.
2. In **ZXAN(config-pon)#** mode, run the **onu-type** command to create an ONU type profile.
3. In **ZXAN(config-pon)#** mode, run the **onu-type-if** command to configure a user port for the ONU type.
4. In **ZXAN(config)#** mode, run the **epon** command to enter EPON configuration mode.
5. In **ZXAN(config-pon)#** mode, run the **auto-dispatch-set** command to disable the automatic delivery function of the ONU type.

– End of Steps –

Example

1. The following example shows how to create an ONU type profile in PON configuration mode:

```
ZXAN(config)#pon
ZXAN(config-pon)#onu-type ZTE-F822A epon speed 10g-asymmetric
ZXAN(config-pon)#onu-type ZTE-F822A epon description 24FE,24POTS
```

2. The following example shows how to configure a user port for the ONU type:

```
ZXAN(config-pon)#onu-type-if ZTE-F822A eth_0/1-24
ZXAN(config-pon)#onu-type-if ZTE-F822A pots_0/1-24
```

3. The following example shows how to disable the automatic delivery function of the ONU type in EPON configuration mode:

```
ZXAN(config)#epon
ZXAN(config-epon)#auto-dispatch-set ZTE-F822A disable
```

4. The following example shows how to display the configuration of an EPON ONU type profile:

```
ZXAN(config-epon)#show onu-type epon ZTE-F822A
ONU type name:          ZTE-F822A
PON type:               epon
Description:            24FE,24POTS
Protect type:           none
```

Speed: 10g-asymmetric

5.2.2 Configuring Authentication Parameters for an EPON ONU

For an EPON ONU that goes online initially, you need to configure authentication parameters for ONU authentication.

Context

The ZXA10 C300/C350/C320 supports the following four ONU authentication modes:

- MAC address authentication (default)
The PON MAC address of an ONU is used for authentication.
- Logical ID authentication
The logical ID of an ONU is used for authentication.
- SN authentication
The serial number of an ONU is used for authentication.
- Hybrid authentication
Any of the MAC address, logical ID, or serial number of an ONU can be used for authentication.

In EPON configuration mode, you can run the **onu-authentication-mode service** command to modify the authentication mode for an ONU.

Steps

1. In **ZXAN(config)#** mode, run the **show epon onu unthentication** command to query an unauthenticated ONU.
2. In **ZXAN(config)#** mode, run the **interface epon-olt** command to enter OLT interface configuration mode.
3. In **ZXAN(config-pon)#** mode, run the **onu type** command to configure ONU authentication parameters.

– End of Steps –

Example

1. The following example shows how to query information about the unauthenticated ONU:

```
ZXAN(config)#show onu unauthentication epon-olt_1/7/1
Onu Interface :    epon-onu_1/7/1:1
Onu Model :       ZTE-F420
MAC Address :     00d0.d09a.46b6
SN :              test1F400MAC0026EDAFCE19
LOID :            test1F400MAC0026EDAFCE19
LOID_PSW :        0026EDAFCE19
```

```

Online State :      authentication deny
ExtOam State :      complete
Software Version :  V1.1.0c_ZL2
Hardware Version :  V1.0.0
RegTime :           2016/10/15 10:03:32

```

- The following example shows how to configure ONU authentication parameters in OLT interface mode:

```

ZXAN(config)#interface epon-olt_1/7/1
ZXAN(config-if)#onu 1 type ZTE-F420 mac 00d0.d09a.46b6
ZXAN(config-if)#exit

```

5.2.3 Configuring an EPON ONU Bandwidth Profile

You can use a bandwidth profile to configure bandwidths for multiple EPON ONUs in batches. The EPON ONUs use the same bandwidth configuration.

Context

The default bandwidth profile on the ZXA10 C300/C350/C320 is named `default`. For its configuration, refer to [Table 5-2](#).

Table 5-2 Default Bandwidth Profile Parameters

Direction	Parameter	Data
Upstream	Maximum bandwidth (MaxBw)	100000 kbps
	Minimum bandwidth (MinBw)	10000 kbps
	Fixed bandwidth (FixBw)	0
	Fixed packet size (FixPktSize)	64 byte
	Maximum burst capacity (MaxBurstSize)	45000 byte
	Priority (Priority)	0
Downstream	Maximum downstream bandwidth (MaxBw)	100000 kbps
	Maximum burst capacity (MaxBurstSize)	16777215 byte

Steps

- In **ZXAN(config-epon)#** mode, run the **onu-sla-profile** command to configure a T-CONT bandwidth profile.
- In **ZXAN(config-epon)#** mode, run the **onu-sla-profile profile-set** command to configure the upstream and downstream bandwidths.

– End of Steps –

Example

1. The following example shows how to create a bandwidth profile in EPON configuration mode:

```
ZXAN(config)#epon
ZXAN(config-epon)#onu-sla-profile profile-name create 10M
```

2. The following example shows how to configure upstream and downstream bandwidths:

```
ZXAN(config-epon)#onu-sla-profile profile-set 10M upstream assured 5000
maximum 10000
ZXAN(config-epon)#onu-sla-profile profile-set 10M downstream maximum 10000
```



Note:

In the upstream bandwidth configuration, it is recommended to configure the assured bandwidth and maximum bandwidth, and use the default values of other parameters.

In the downstream bandwidth configuration, the maximum downstream bandwidth must be configured. The system automatically calculates and sets the maximum burst packet parameter in accordance with the maximum downstream bandwidth.

5.2.4 Configuring an EPON VoIP IP Profile

By using an EPON IP profile, you can configure IP addresses for ONUs in batches.

Context

The ZXA10 C300/C350/C320 supports the following three IP address assignment modes:

- Static
- DHCP
- PPPoE

One ONU can use only one IP address assignment mode.

The VoIP IP profile is applicable to only the static assignment mode.

Steps

1. In **ZXAN(config-epon)#** mode, run the **voip-ip profile** command to configure a VoIP IP profile.

– End of Steps –

Example

The following example shows how to configure a VoIP IP profile in EPON configuration mode:

```
ZXAN(config)#epon
ZXAN(config-epon)#voip-ip profile ip-test relation independent mode static gateway
1.2.1.1 dns 1.1.1.1
```

5.2.5 Configuring an EPON VoIP VLAN Profile

By using an [EPON VoIP VLAN](#) profile, you can configure VoIP VLANs for EPON ONUs in batches.

Steps

1. In **ZXAN(config-epon)#** mode, run the **voip-vlan profile** command to configure a VoIP VLAN profile.

– End of Steps –

Example

The following example shows how to configure a VoIP VLAN profile in EPON configuration mode:

```
ZXAN(config)#epon
ZXAN(config-epon)#voip-vlan profile vlan vlan-test tag-mode tag cvlan 300 priority 7
```

5.2.6 Configuring an EPON H.248 Profile

By using an EPON H.248 profile, you can configure H.248 parameters for EPON ONUs in batches.

Steps

1. In **ZXAN(config)#** mode, run the **epon** command to enter EPON configuration mode.
2. In **ZXAN(config-epon)#** mode, run the **h248-profile** command to configure an H.248 profile.

– End of Steps –

Example

The following example shows how to configure an H.248 profile in EPON configuration mode:

```
ZXAN(config)#epon
ZXAN(config-epon)#h248-profile h248-test register-server ip 1.2.3.1
ZXAN(config-epon)#h248-profile h248-test backup-register-server ip 1.2.3.2
ZXAN(config-epon)#h248-profile h248-test mg register-mode domainname
```

5.2.7 Configuring an EPON MGCP Profile

By using an EPON MGCP profile, you can configure MGCP parameters for EPON ONUs in batches.

Steps

1. In **ZXAN(config)#** mode, run the **epon** command to enter EPON configuration mode.
2. In **ZXAN(config-epon)#** mode, run the **mgcp-profile** command to configure an MGCP profile.

– End of Steps –

Example

The following example shows how to configure an MGCP profile in EPON configuration mode:

```
ZXAN(config)#epon
ZXAN(config-epon)#mgcp-profile mgcp-test register-server ip 1.2.3.1
ZXAN(config-epon)#mgcp-profile mgcp-test backup-register-server ip 1.2.3.2
ZXAN(config-epon)#mgcp-profile mgcp-test mg register-mode domainname
```

5.2.8 Configuring an EPON SIP Profile

By using an EPON SIP profile, you can configure SIP parameters for EPON ONUs in batches.

Steps

1. In **ZXAN(config)#** mode, run the **epon** command to enter EPON configuration mode.
2. In **ZXAN(config-epon)#** mode, run the **sip-profile** command to configure a SIP profile.

– End of Steps –

Example

The following example shows how to configure a SIP profile in EPON configuration mode:

```
ZXAN(config)#epon
ZXAN(config-epon)#sip-profile sip-test register-server ip 1.2.3.1
ZXAN(config-epon)#sip-profile sip-test backup-register-server ip 1.2.3.2
ZXAN(config-epon)#sip-profile sip-test proxy-server ip 1.2.3.1
ZXAN(config-epon)#sip-profile sip-test backup-proxy-server ip 1.2.3.2
ZXAN(config-epon)#sip-profile sip-test outbound-server ip 1.2.3.1
```

5.2.9 Configuring the EPON Broadband Service

After you configure the EPON broadband service, subscribers can access the Internet.

Prerequisite

- The EPON ONU is authenticated.
- The EPON ONU bandwidth profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface epon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, configure the ONU bandwidth.
 - Run the **sla-profile** command to apply a bandwidth profile to the ONU.
 - Run the **sla** command to modify ONU bandwidth.
3. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
4. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a service port VLAN.
6. In **ZXAN(config)#** mode, run the **pon-onu-mng epon-onu** command to enter remote ONU management mode.
7. In **ZXAN(epon-onu-mng)#** mode, run the **vlan port** command to configure an ONU user port VLAN.

– End of Steps –

Example

1. The following example shows how to configure the ONU bandwidth in ONU interface mode:
 - Applying a bandwidth profile to the ONU:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#sla-profile 10M
ZXAN(config-if)#exit
```
 - Modify the ONU bandwidth:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#sla upstream assured 5000 maximum 10000
ZXAN(config-if)#sla downstream maximum 10000
ZXAN(config-if)#exit
```
2. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 101 tag
ZXAN(config-if)#exit
```
3. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#switchport vlan 101 tag
ZXAN(config-if)#exit
```

4. The following example shows how to configure a user port VLAN in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng epon-onu_1/7/1:1
ZXAN(epon-onu-mng)#vlan port eth_0/1 mode tag vlan 101 priority 0
ZXAN(epon-onu-mng)#end
```

5. Save the configurations.

```
ZXAN#write
```

5.2.10 Configuring the EPON Multicast Service

After you configure the EPON multicast service, subscribers can receive multicast service streams.

Prerequisite

- The EPON ONU is authenticated.
- The EPON ONU bandwidth profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface epon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **sla-profile** command to configure the ONU bandwidth.
3. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
4. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a service port VLAN.
6. In **ZXAN(config)#** mode, run the **igmp enable** command to enable the IGMP protocol globally.
7. In **ZXAN(config-if)#** mode, run the **igmp fast-leave** command to configure the port IGMP parameters for the ONU.
8. In **ZXAN(config-if)#** mode, run the **igmp mvlan** command to configure an MVLAN.
9. In **ZXAN(config-if)#** mode, run the **igmp mvlan group** command to configure an MVLAN multicast group.
10. In **ZXAN(config-if)#** mode, run the **igmp mvlan source-port** command to configure an MVLAN source port.

11. In **ZXAN(config-if)#** mode, run the **igmp mvlan receive-port** command to configure an MVLAN receiving port.
12. In **ZXAN(epon-onu-mng)#** mode, run the **multicast vlan** and **multicast vlan tag-strip** commands to configure an ONU user port MVLAN.
13. In **ZXAN(epon-onu-mng)#** mode, run the **vlan port** command to configure an ONU user port VLAN.
14. In **ZXAN#** mode, run the **write** command to save the configuration data.

– End of Steps –

Example

1. The following example shows how to configure the ONU bandwidth in ONU interface mode:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#sla-profile 10M
ZXAN(config-if)#exit
```

2. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 201 tag
ZXAN(config-if)#exit
```

3. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#switchport vlan 201 tag
ZXAN(config-if)#exit
```

4. (Optional) The following example shows how to enable the IGMP protocol globally:

```
ZXAN(config)#igmp enable
```

5. The following example shows how to configure the port IGMP parameters:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#igmp fast-leave enable
ZXAN(config-if)#exit
```

6. The following example shows how to configure an MVLAN:

```
ZXAN(config)#igmp mvlan 201
```

7. (Optional) The following example shows how to configure the MVLAN operating mode:

```
ZXAN(config)#igmp mvlan 201 work-mode proxy
```

8. The following example shows how to configure an MVLAN multicast group:

```
ZXAN(config)#igmp mvlan 201 group 224.1.1.1 to 224.1.1.3
```

9. The following example shows how to configure an MVLAN source port:

```
ZXAN(config)#igmp mvlan 201 source-port gei_1/20/1
```

10. The following example shows how to configure an MVLAN receiving port:

```
ZXAN(config)#igmp mvlan 201 receive-port epon-onu_1/7/1:1
```

11. The following example shows how to configure a user port MVLAN in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng epon-onu_1/7/1:1
ZXAN(epon-onu-mng)#multicast vlan port eth_0/2 add vlanlist 201
ZXAN(epon-onu-mng)#multicast vlan tag-strip port eth_0/2 enable
```

12. The following example shows how to configure a user port VLAN:

```
ZXAN(epon-onu-mng)#vlan port eth_0/2 mode tag vlan 201 priority 5
ZXAN(epon-onu-mng)#end
```

13. Save the configurations.

```
ZXAN#write
```

5.2.11 Configuring the EPON Voice Service (SIP)

After you configure the EPON voice service, subscribers can make and answer phone calls.

Prerequisite

- The EPON ONU is authenticated.
- The EPON ONU bandwidth profile is configured.
- The EPON VoIP IP profile is configured.
- The EPON VoIP VLAN profile is configured.
- The EPON SIP profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface epon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **sla-profile** command to configure the ONU bandwidth.
3. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
4. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a service port VLAN.
6. In **ZXAN(epon-onu-mng)#** mode, run the **voip-module global-profile** command to configure a global VoIP profile.
7. In **ZXAN(epon-onu-mng)#** mode, run the **voip-module protocol-profile** command to configure the VoIP protocol type.

8. In **ZXAN(epon-onu-mng)#** mode, run the **voip ip-address** command to configure an MGIP address.
 9. In **ZXAN(epon-onu-mng)#** mode, run the **voip sip-user** command to configure a SIP user.
 10. Save the configurations.
- End of Steps –**

Example

1. The following example shows how to configure the ONU bandwidth in ONU interface mode:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#sla-profile 10M
ZXAN(config-if)#exit
```

2. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 301 tag
ZXAN(config-if)#exit
```

3. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#switchport vlan 301 tag
ZXAN(config-if)#exit
```

4. The following example shows how to configure a global VoIP profile in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng epon-onu_1/7/1:1
ZXAN(epon-onu-mng)#voip-module global-profile apply ip ip-test vlan vlan-test
```

5. The following example shows how to configure a VoIP protocol profile:

```
ZXAN(epon-onu-mng)#voip-module protocol-profile apply sip sip-test
```

6. The following example shows how to configuring the MG IP address:

```
ZXAN(epon-onu-mng)#voip ip-address 1.2.3.4 mask 255.255.255.0 slot 1
```

7. The following example shows how to configure a SIP user:

```
ZXAN(epon-onu-mng)#voip sip-user account 12345 name 12345 password 12345 interface
pots_0/1
ZXAN(epon-onu-mng)#end
```

8. Save the configurations.

```
ZXAN#write
```

5.2.12 Configuring the EPON Voice Service (H.248)

After you configure the EPON voice service, subscribers can make and answer phone calls.

Prerequisite

- The EPON ONU is authenticated.
- The EPON ONU bandwidth profile is configured.
- The EPON VoIP IP profile is configured.
- The EPON VoIP VLAN profile is configured.
- The EPON H.248 profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface epon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **sla-profile** command to configure the ONU bandwidth.
3. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
4. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a service port VLAN.
6. In **ZXAN(epon-onu-mng)#** mode, run the **voip-module global-profile** command to configure a global VoIP profile.
7. In **ZXAN(epon-onu-mng)#** mode, run the **voip mg-domainname** command to configure the VoIP protocol type.
8. In **ZXAN(epon-onu-mng)#** mode, run the **voip ip-address** command to configure an MGIP address.
9. In **ZXAN(epon-onu-mng)#** mode, run the **voip-module protocol-profile** command to configure a VoIP protocol profile.
10. In **ZXAN(epon-onu-mng)#** mode, run the **voip user-tid** command to configure a user TID.
11. In **ZXAN(epon-onu-mng)#** mode, run the **voip rtp-tid** command to configure an RTP TID.
12. Save the configurations.

– End of Steps –

Example

1. The following example shows how to configure the ONU bandwidth in ONU interface mode:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#sla-profile 10M
ZXAN(config-if)#exit
```

2. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 301 tag
ZXAN(config-if)#exit
```

3. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface epon-onu_1/7/1:1
ZXAN(config-if)#switchport vlan 301 tag
ZXAN(config-if)#exit
```

4. The following example shows how to configure a global VoIP profile in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng epon-onu_1/7/1:1
ZXAN(epon-onu-mng)#voip-module global-profile apply ip ip-test vlan vlan-test
```

5. The following example shows how to configure the MG domain name:

```
ZXAN(epon-onu-mng)#voip mg-domainname iad.zte.com.cn slot 1
```

6. The following example shows how to configure the MG IP address:

```
ZXAN(epon-onu-mng)#voip ip-address 1.2.3.4 mask 255.255.255.0 slot 1
```

7. The following example shows how to configure a VoIP protocol profile:

```
ZXAN(epon-onu-mng)#voip-module protocol-profile apply h248 h248-test
```

8. The following example shows how to configure a user TID:

```
ZXAN(epon-onu-mng)#voip user-tid 1 beginid 1 tid-num 2 prefix USER begin-digit 0
align enable digit-length 2 slot 1
```

9. The following example shows how to configure an RTP TID:

```
ZXAN(epon-onu-mng)#voip rtp-tid prefix RTP begin-digit 0 align enable digit-length
2 slot 1
ZXAN(epon-onu-mng)#end
```

10. Save the configurations.

```
ZXAN#write
```

5.2.13 Configuring the EPON P2P Service

After you configure the Peer-to-Peer ([P2P](#)) service, ONU service interworking is implemented under the EPON OLT interface.

Prerequisite

The EPON broadband, multicast, or voice service is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface epon-olt** command to enter OLT interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **p2p mode group**, **p2p group**, and **p2p group-name** commands to configure the P2P mode and a P2P group.
3. In **ZXAN(config)#** mode, run the **interface epon-onu** command to enter ONU interface configuration mode.
4. In **ZXAN(config-if)#** mode (ONU interface configuration mode), run the **join-p2p-group p** command to add the ONU to the P2P group.
5. Save the configurations.

– End of Steps –

Example

1. The following example shows how to configure the P2P mode and a P2P group in OLT interface configuration mode:

```
ZXAN(config)#interface epon-olt_1/3/1
ZXAN(config-if)#p2p mode group
ZXAN(config-if)#p2p group 1
ZXAN(config-if)#p2p group-name 1 test-group
```

2. The following example shows how to add the ONU to the P2P group in ONU interface configuration mode:

```
ZXAN(config)#interface epon-onu_1/3/1:1
ZXAN(config-if)#join-p2p-group 1
ZXAN(config-if)#exit
ZXAN(config)#interface epon-onu_1/3/1:2
ZXAN(config-if)#join-p2p-group 1
```

5.2.14 Configuring the 10G-EPON Service

The ZXA10 C300/C350/C320 is connected to an ONU device through the 10G-EPON interface to provide Internet, video, and voice services to subscribers.

Context

The ZXA10 C300/C350/C320 supports three modes of 10G-EPON services:

- 10G symmetric mode
- 10G asymmetric mode
- 1G mode (default)

Steps

Configure the ONU

1. In **ZXAN(config-if)#** mode, run the **speed** command to configure OLT work mode.
2. In **ZXAN(config-if)#** mode, run the **onu type** command to configure ONU authentication parameters.
3. In **ZXAN(config-if)#** mode, run the **admin enable** command to enable ONU authentication protocol.
4. In **ZXAN(config-if)#** mode, run the **sla** command to configure ONU bandwidth.

Configure the VLAN

5. In **ZXAN(config-if)#** mode, run the **switchport** command to configure uplink interface VLAN.
6. In **ZXAN(config-if)#** mode, run the **service-port** command to configure ONU interface VLAN.

Configure the MVLAN

7. In **ZXAN(config)#** mode, run the **igmp enable** command to enable global IGMP.
8. In **ZXAN(config-if)#** mode, run the **igmp fast-leave** command to configure ONU interface IGMP parameters.
9. In **ZXAN(config)#** mode, run the **igmp mvlan** command to configure MVLAN.
10. In **ZXAN(config)#** mode, run the **igmp mvlan group** command to configure MVLAN groups.
11. In **ZXAN(config)#** mode, run the **igmp mvlan source-port** command to configure MVLAN source port.
12. In **ZXAN(config)#** mode, run the **igmp mvlan receive-port** command to configure MVLAN receiving port.

Configure the ONU multicast service

13. In **ZXAN(epon-onu-mng)#** mode, run the **multicast vlan** command to configure user port MVLAN.
14. In **ZXAN(epon-onu-mng)#** mode, run the **multicast vlan tag-strip** command to enable MVLAN remove function on the user port.
15. In **ZXAN(epon-onu-mng)#** mode, run the **vlan port** command to configure the user port VLAN.

Configure the ONU management IP address

16. In **ZXAN(epon-onu-mng)#** mode, run the **mgmt-ip** command to configure ONU management IP address.

– End of Steps –

Example

1. The following example shows how to configure the 10G-EPON operating mode in OLT interface configuration mode:

```
ZXAN(config)#interface epon-olt_1/17/1
```

```
ZXAN(config-if)#speed 10g-asymmetric
```

2. The following example shows how to authenticate an ONU:

```
ZXAN(config-if)#onu 1 type ZTE-F822A mac 0019.c600.0011
ZXAN(config-if)#exit
```

3. The following example shows how to enable the authentication protocol for the ONU in ONU interface configuration mode:

```
ZXAN(config)#interface epon-onu_1/17/1:1
ZXAN(config-if)#admin enable
```

4. The following example shows how to configure the ONU bandwidths:

```
ZXAN(config-if)#sla upstream assured 10000 maximum 100000
ZXAN(config-if)#sla downstream maximum 100000
ZXAN(config-if)#exit
```

5. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 110,120,130,1000 tag
ZXAN(config-if)#exit
```

6. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface epon-onu_1/17/1:1
ZXAN(config-if)#service-port 1 user-vlan 110 vlan 110
ZXAN(config-if)#service-port 2 user-vlan 120 vlan 120
ZXAN(config-if)#service-port 3 user-vlan 130 vlan 130
ZXAN(config-if)#service-port 4 user-vlan 1000 vlan 1000
ZXAN(config-if)#exit
```

7. (Optional) The following example shows how to enable the IGMP protocol globally:

```
ZXAN(config)#igmp enable
```

8. The following example shows how to configure the port IGMP parameters:

```
ZXAN(config)#interface epon-onu_1/17/1:1
ZXAN(config-if)#igmp fast-leave enable
ZXAN(config-if)#exit
```

9. The following example shows how to configure an MVLAN:

```
ZXAN(config)#igmp mvlan 120
ZXAN(config)#igmp mvlan 120 work-mode proxy
ZXAN(config)#igmp mvlan 120 group 224.1.1.1 to 224.1.1.3
ZXAN(config)#igmp mvlan 120 source-port gei_1/20/1
ZXAN(config)#igmp mvlan 120 receive-port epon-onu_1/17/1:1
```

10. The following example shows how to configure the ONU broadband service in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng epon-onu_1/17/1:1
```

```
ZXAN(epon-onu-mng)#vlan port eth_0/1 mode tag vlan 110 priority 0
```

11. The following example shows how to configure the ONU multicast service:

```
ZXAN(epon-onu-mng)#multicast switch igmpsnooping
ZXAN(epon-onu-mng)#multicast fastleave enable
ZXAN(epon-onu-mng)#vlan port eth_0/2 mode tag vlan 120 priority 5
ZXAN(epon-onu-mng)#multicast vlan port eth_0/2 add vlanlist 120
ZXAN(epon-onu-mng)#multicast vlan tag-strip port eth_0/2 enable
```

12. The following example shows how to configure the management IP address for the ONU:

```
ZXAN(epon-onu-mng)#mgmt-ip 10.1.1.1 255.255.255.0 vlan 1000 priority 7 route
0.0.0.0 0.0.0.0 10.1.1.254 status enable
```

13. Log in to the F822 device and configure the voice service.

- a. The following example shows how to configure a voice service VLAN:

```
ZXAN(config)#vlan 300
ZXAN(config-vlan)#exit
ZXAN(config)#interface gei_0/3/1
ZXAN(config-if)#switchport vlan 300 tag
ZXAN(config-if)#exit
```

- b. The following example shows how to the IP address and route for the voice service:

```
ZXAN(config)#interface inband-vlan 300
ZXAN(config-if)#ip address 10.2.2.2 255.255.255.0 voip
ZXAN(config-if)#exit
ZXAN(config)#ip route 0.0.0.0 0.0.0.0 10.2.2.1 name zteroute
ZXAN(config)#exit
```

- c. The following example shows how to configure the MGC:

```
ZXAN#ag
ZXAN(ag)#mod-mgc mgcid 1 mgcip 2.1.1.2 mgctype 1
```

- d. (Optional) The following example shows how to configure an SLC TID and an IPS TID:

```
ZXAN(ag)#del-slctermid beginslot 1 num 48
ZXAN(ag)#add-slctermid beginslot 1 num 48 tmidfix USER type 3 digitlen 3 mgid 1
ZXAN(ag)#del-ipstermid
ZXAN(ag)#add-ipstermid tmidfix RTP type 3 digitlen 3 mgid 1
```

14. Save the configurations.

```
ZXAN#write
```

5.3 GPON/XG-PON Service Configuration

GPON access is a flexible access technology that provides super bandwidth access in both broadband and narrowband service environments. It supports multiple rate modes and uses a single optical fiber to provide a user with the voice, data, and video services.

The principles and configuration of the [XG-PON](#) services and the GPON services are the same except that the XG-PON provides a data bandwidth of 10 Gbps.

5.3.1 Configuring a GPON ONU Type Profile

Before authentication of a [GPON](#) Optical Network Unit (ONU), you need to create an ONU type profile if an ONU type does not exist.

Context

The ZXA10 C300/C350/C320 supports the following default GPON [ONU](#) types:

- ZTE-F601
- ZTE-F621
- ZTE-F622
- ZTE-F625
- ZTE-F628
- ZTE-F640
- ZTE-F641

On the ZXA10 C300/C350/C320, ZTE-9806, ZTE-F822, and ZTE-F820 are EPON ONU types. If you need to configure the corresponding GPON ONU types, you can use ZTEG-9806H, ZTEG-F822, and ZTEG-F820 respectively.

You can use the **show onu-type gpon** command to query the default GPON ONU types.

Steps

1. In **ZXAN(config)#** mode, run the **pon** command to enter PON configuration mode.
2. In **ZXAN(config-pon)#** mode, run the **onu-type** command to create an ONU type profile.
3. In **ZXAN(config-pon)#** mode, run the **onu-type-if** command to configure a user port for the ONU type.

– End of Steps –

Example

1. The following example shows how to create an ONU type profile in PON configuration mode:

```
ZXAN(config)#pon
ZXAN(config-pon)#onu-type ZTEG-F620 gpon description 4FE,2POTS max-tcont 7
max-gemport 32 max-switch-perslot 1 max-flow-perswitch 32
```

2. The following example shows how to configure a user port for the ONU type:

```
ZXAN(config-pon)#onu-type-if ZTEG-F620 eth_0/1-4
ZXAN(config-pon)#onu-type-if ZTEG-F620 pots_0/1-2
```

3. (Optional) The following example shows how to display the configured ONU type profile:

```
ZXAN(config-pon)#show onu-type gpon ZTEG-F620
ONU type name:          ZTEG-F620
PON type:               gpon
Description:             4FE,2POTS
Max T-CONT:             7
Max GEM port:           32
Max switch per slot:    1
Max flow per switch:    32
Max IP host:            2
Max IPv6 host:          0
Service ability N:1:    support
Service ability 1:M:    support
Service ability 1:P:    support
WIFI mgmt via non OMCI: disable
OMCI send mode:         async
Default multicast range:none
VRG:                    disable
MGC configure mode:     zte
Max VEIP:               0(default: 1 VEIP)
Extended OMCI:          disable
```

5.3.2 Configuring Authentication Parameters for a GPON ONU

For a [GPON ONU](#) that goes online initially, you need to configure authentication parameters for ONU authentication.

Context

For a description of the ONU phase states, refer to [Table 5-3](#).

Table 5-3 ONU Phase State Descriptions

State	Description
Offline	The OLT does not find the ONU because the ONU is offline.
Logging	The OLT has found the ONU and is measuring the distance.
syncMib	The OLT has measured the distance to the ONU and is synchronizing data.
Working	The data synchronization between the OLT and the ONU is completed, and you can configure services.
LOS	The fiber link between the OLT and the ONU is faulty and the ONU is offline.
DyingGasp	The ONU is powered off.

Steps

1. In **ZXAN(config)#** mode, run the **show gpon onu uncfg gpon-olt** command to query an unauthenticated ONU.
2. In **ZXAN(config)#** mode, run the **interface gpon-olt** command to enter OLT interface configuration mode.
3. In **ZXAN(config-gpon)#** mode, run the **onu type** command to configure ONU authentication parameters.

– End of Steps –

Example

1. The following example shows how to query information about the unauthenticated ONU:

```
ZXAN(config)#show gpon onu uncfg gpon-olt_1/5/1
OnuIndex          Sn          State
-----
gpon-onu_1/5/1:1  ZTEG00000002  unknown
```

2. The following example shows how to configure ONU authentication parameters in OLT interface mode:

```
ZXAN(config)#interface gpon-olt_1/5/1
ZXAN(config-if)#onu 1 type ZTEG-F620 sn ZTEG000000002
[Successful]
ZXAN(config-if)#exit
```

3. (Optional) The following example shows how to query information about the authenticated ONU:

```
ZXAN(config)#show gpon onu state gpon-olt_1/5/1
OnuIndex  Admin State  OMCC State  Phase State  Config State  Channel
-----
1/5/1:1   enable       enable      Working      succeeded      1 (GPON)
ONU Number: 0/1
```

5.3.3 Configuring a T-CONT Bandwidth Profile

The T-CONT bandwidth profile describes the Transmission Container (T-CONT) traffic parameters. By specifying a T-CONT bandwidth profile, you can implement T-CONT traffic control.

Context

The ZXA10 C300/C350/C320 supports a maximum of 512 T-CONT profiles.

There are the following five types of upstream bandwidths:

- Fixed bandwidth
- Assured bandwidth

- Non-assured bandwidth
- Best-effort bandwidth
- Maximum bandwidth

The priorities of fixed bandwidth, assured bandwidth, non-assured bandwidth, best-effort bandwidth, and maximum bandwidth are in descending order. A T-CONT bandwidth profile may contains one or multiple types of bandwidths. Five types of T-CONT bandwidth profiles are as follows:

- Fixed bandwidth (type 1)

Type 1 includes only fixed bandwidth. Type 1 has fixed bandwidth and timeslots. It is applicable to any service that is sensitive to delay and jitter and has fixed or stable flow rate, such as the voice service.

- Assured bandwidth (type 2)

Type 2 includes only assured bandwidth. Type 2 has assured bandwidth but uncertain timeslots. It is applicable to any service that is insensitive to delay and jitter and has a limited flow rate, such as the Video On Demand (VOD) service.

- Assured and non-assured bandwidths (type 3)

Type 3 includes assured and non-assured bandwidths. It has the minimum assured bandwidth and shares the excess bandwidth dynamically. Meanwhile, it is constrained by the maximum bandwidth. It is applicable to any service that requires service assurance but has traffic surges, such as the subscription download service.

- Best-effort bandwidth (type 4)

Type 4 includes only best-effort bandwidth. After the fixed bandwidth, assured bandwidth, and non-assured bandwidth are allocated, the excess bandwidth is competed for. It is applicable to any service that is insensitive to delay and jitter, such as the Web browsing service.

- Support all (type 5)

Type 5 includes the four types and has the characteristics of the four types. It is applicable to most service streams. The sum of fixed bandwidth and assured bandwidth on a PON port must be no more than 1 Gbps.

Steps

1. In **ZXAN(config)#** mode, run the **profile tcont** command to configure a T-CONT bandwidth profile.

– End of Steps –

Example

1. The following example shows how to create a T-CONT bandwidth profile in GPON configuration mode:

```
ZXAN(config)#gpon
ZXAN(config-gpon)#profile tcont 20M type 5 fixed 2000 assured 5000 maximum 20000
```

```
[Successful]
ZXAN(config-gpon)#profile tcont 15M type 4 maximum 15000
[Successful]
ZXAN(config-gpon)#profile tcont 10M type 3 assured 5000 maximum 10000
[Successful]
ZXAN(config-gpon)#profile tcont 5M type 2 assured 5000
[Successful]
ZXAN(config-gpon)#profile tcont 2M type 1 fixed 2000
[Successful]
```

2. (Optional) The following example shows how to display T-CONT bandwidth profiles:

```
ZXAN(config-gpon)#show gpon profile tcont
Profile name :default
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)  PRIORITY  WEIGHT
1             10000      0          0          N/A       N/A

Profile name :20M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)  PRIORITY  WEIGHT
5             2000       5000       20000      N/A       N/A

Profile name :15M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)  PRIORITY  WEIGHT
4             0          0          15000      N/A       N/A

Profile name :10M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)  PRIORITY  WEIGHT
3             0          5000       10000      N/A       N/A

Profile name :5M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)  PRIORITY  WEIGHT
2             0          5000       0          N/A       N/A

Profile name :2M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)  PRIORITY  WEIGHT
1             2000       0          0          N/A       N/A
```

5.3.4 Configuring a GPON ONU IP Profile

By using a [GPON ONU](#) IP profile, you can configure IP addresses for GPON ONUs in batches.

Context

The ZXA10 C300/C350/C320 supports the following three IP address assignment modes:

- Static
- [DHCP](#)

- **PPPoE**

One ONU can use only one IP address assignment mode.

The GPON ONU IP profile is applicable to only the static assignment mode.

Steps

1. In **ZXAN(config)#** mode, run the **gpon** command to enter GPON configuration mode.
2. In **ZXAN(config-gpon)#** mode, run the **onu profile ip** command to configure a GPON ONU IP profile.

– End of Steps –

Example

1. The following example shows how to configure a GPON ONU IP profile in GPON configuration mode:

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile ip ip-test gateway 1.2.3.1 primary-dns 1.1.1.1
```

2. (Optional) The following example shows how to display a GPON ONU IP profile:

```
ZXAN(config-gpon)#show gpon onu profile ip ip-test
Profile name:  ip-test
Gateway:       1.2.3.1
Primary DNS:   1.1.1.1
Secondary DNS: 0.0.0.0
```

5.3.5 Configuring a GPON ONU VLAN Profile

By using a **GPON ONU VLAN** profile, you can configure VLANs for GPON ONUs in batches.

Steps

1. In **ZXAN(config)#** mode, run the **gpon** command to enter GPON configuration mode.
2. In **ZXAN(config-gpon)#** mode, run the **onu profile vlan** command to configure a GPON ONU VLAN profile.

– End of Steps –

Example

1. The following example shows how to configure a GPON ONU VLAN profile in GPON configuration mode:

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile vlan vlan-test tag-mode tag cvlan 300 pri 7
```

2. (Optional) The following example shows how to display a GPON ONU VLAN profile:

```
ZXAN(config-gpon)#show gpon onu profile vlan vlan-test
```

```

Profile name:  vlan-test
Tag mode:      tag
CVLAN:        300
CVLAN priority:7

```

5.3.6 Configuring a GPON VoIP Access Code Profile

By using a **VoIP** access code profile, you can configure access codes for the VoIP supplementary services in batches on the GPON ONU. The VoIP service is based on the **SIP**.

Context

You can set the relationships between access codes and corresponding services on ONUs by configuring the access code of each supplementary service. When a subscriber dials an access code, the corresponding service is activated on the ONU (SIP agent), and then processed in accordance with the service procedure.

Steps

1. In **ZXAN(config)#** mode, run the **gpon** command to enter GPON configuration mode.
2. In **ZXAN(config-gpon)#** mode, run the **onu profile voip-accesscode** command to configure a VoIP access code profile.

– End of Steps –

Example

1. The following example shows how to configure a VoIP access code profile in GPON configuration mode:

```

ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile voip-accesscode abc call-hold ***

```

2. (Optional) The following example shows how to display a VoIP access code profile:

```

ZXAN(config-gpon)#show gpon onu profile voip-accesscode abc
Profile name:                               abc
Cancel call waiting:
Call hold:                                  ***
Call park:
CID activate:
CID deactivate:
No disturb activation:
No disturb deactivation:
No disturb pin change:
Emergency service num:
Intercom service:
Unattend blind call transfer:

```

Attend call transfer:

5.3.7 Configuring a GPON VoIP Service Application Profile

By using a **VoIP** service application profile, you can configure the **SIP**-based VoIP supplementary services for GPON ONUs in batches.

Steps

1. In **ZXAN(config)#** mode, run the **gpon** command to enter GPON configuration mode.
2. In **ZXAN(config-gpon)#** mode, run the **onu profile voip-appsrv voip-service** command to configure a VoIP service application profile.

– End of Steps –

Example

1. The following example shows how to configure a VoIP service application profile in GPON configuration mode:

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile voip-appsrv voip-service call-waiting enable call-transfer
enable call-hold enable 3way enable
```

2. (Optional) The following example shows how to display a VoIP service application profile:

```
ZXAN(config-gpon)#show gpon onu profile voip-appsrv voip-service
Profile name:                voip-service
Calling num:                 disable
Calling name:                disable
CID blocking:                disable
CID num permanent status:    disable
CID name permanent status:   disable
Anonymous CID blocking:      disable
Call waiting:                enable
CID announcement:            disable
3way:                        enable
Call transfer:                enable
Call hold:                   enable
Call park:                   disable
No disturb:                  disable
Emergency call flash:         disable
Emergency originate hold:     disable
6way:                        disable
Message waiting splash ring:  disable
Message waiting special dial tone: disable
Message waiting visual indication: disable
Call forwarding ind:          disable
```

```

Direct connect feature:          disable
Dialtone delay:                 disable
Direct connect URI:
    Validation scheme:          disable
    Username:
    Password:
    Realm:
Bridge line agent URI:
    Validation scheme:          disable
    Username:
    Password:
    Realm:
Conference factory URI:
    Validation scheme:          disable
    Username:
    Password:
    Realm:

```

5.3.8 Configuring a SIP Profile

By using a GPON SIP profile, you can configure [SIP](#) parameters for GPON ONUs in batches.

Prerequisite

- The GPON VoIP access code profile is configured.
- The GPON VoIP service application profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **gpon** command to enter GPON configuration mode.
2. In **ZXAN(config-gpon)#** mode, run the **onu profile sip** command to configure a SIP profile.

– End of Steps –

Example

1. The following example shows how to configure a SIP profile in GPON configuration mode:

```

ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile sip sip-test proxy 1.2.3.1
ZXAN(config-gpon)#onu profile sip sip-test accesscode abc
ZXAN(config-gpon)#onu profile sip sip-test appsrv voip-service

```

2. (Optional) The following example shows how to display a SIP profile:

```

ZXAN(config-gpon)#show gpon onu profile sip sip-test

```

```

Profile name :                sip-test
Proxy server:                1.2.3.1
Outbound proxy:              1.2.3.1
Registrar:                  1.2.3.1
    Validation scheme:        disable
UDP port:                   5060
DSCP/TOS:                   0
Media UDP port:              5060
Media DSCP/TOS:              0
DNS1:                       0.0.0.0
DNS2:                       0.0.0.0
Registration expiration time: 3600(s)
Re-registration time:        360(s)
User domain:
Softswitch vendor:
Dial plan table name:        N/A
Release timer:               10(s)
ROH timer:                   15(s)
Link test:                   disable
Link test interval:          N/A
Appsrv:                      voip-service
Accesscode:                  abc

```

5.3.9 Configuring an MGC Profile

By using a GPON Media Gateway Controller ([MGC](#)) profile, you can configure MGC parameters for GPON ONUs in batches.

Context

The ZXA10 C300/C350/C320 supports the following two MGC protocols:

- MGCP
- H.248

Steps

1. In **ZXAN(config)#** mode, run the **gpon** command to enter GPON configuration mode.
2. In **ZXAN(config-gpon)#** mode, run the **onu profile mgc server** command to configure the active and standby MGC servers.
3. In **ZXAN(config-gpon)#** mode, run the **onu profile mgc user-tid** command to configure a user line Terminal Identification ([TID](#)), and the **onu profile mgc rtp-tid** command to configure a Real-time Transport Protocol ([RTP](#)) TID.

– End of Steps –

Example

1. The following example shows how to configure the active and standby MGC servers:

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile mgc mgc-test server1 1.2.3.1
ZXAN(config-gpon)#onu profile mgc mgc-test server2 1.2.3.2
```

2. The following example shows how to configure a user line TID and an RTP TID:

```
ZXAN(config-gpon)#onu profile mgc mgc-test user-tid prefix user postfix-len 5
postfix-start 1
ZXAN(config-gpon)#onu profile mgc mgc-test rtp-tid prefix rtp postfix-len 5
postfix-start 1
```

3. (Optional) The following example shows how to display MGC profile:

```
ZXAN(config-gpon)#show gpon onu profile mgc mgc-test
```

```
Profile name:          mgc-test
Server1:              1.2.3.1
  Validation scheme:   disable
  Username:           N/A
  Password:           N/A
  Realm:              N/A
Server2:              1.2.3.2
  Validation scheme:   disable
  Username:           N/A
  Password:           N/A
  Realm:              N/A
UDP port:             2944
DSCP/TOS:             0
Media UDP port:       2944
Media DSCP/TOS:       0
Message format:       text long
Version:              1
Maximum retry time:   0(s)
Maximum retry attempts: 0(s)
Service change delay: 0(s)
Softswitch vendor:
User TID:
  Prefix:             user
  Postfix length:     5
  Postfix start number: 1
RTP TID:
  Prefix:             rtp
  Postfix length:     5
  Postfix start number: 1
Heart beat:           service change
RTP link detect:      disable
```

```

Number shortest match:  disable
Digit map long timer:   20000 (ms)
Digit map short timer:  5000 (ms)
Digit map start timer:  10000 (ms)
Heart beat interval:    60000 (ms)
Rereg timer min:        60000 (ms)
Rereg timer max:        120000 (ms)
Regmsg retrans timer:   2000 (ms)
Total retrans timer:    10000 (ms)

```

5.3.10 Configuring the GPON Broadband Service

After you configure the GPON broadband service, subscribers can access the Internet.

Prerequisite

- The GPON ONU is authenticated.
- The T-CONT broadband profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface gpon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **tcont** command to configure the T-CONT.
3. In **ZXAN(config-if)#** mode, run the **gemport** command to configure a GEM port.
4. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.
6. In **ZXAN(config-if)#** mode, run the **service-port** command to configure a service port VLAN.
7. In **ZXAN(config)#** mode, run the **pon-onu-mng gpon-onu** command to enter ONU remote management mode.
8. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **service** command to configure a service channel.
9. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **vlan port** command to configure an ONU user port VLAN.

– End of Steps –

Example

1. The following example shows how to configure the T-CONT in ONU interface mode:

```

ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 1 name T1 profile 10M

```

2. The following example shows how to configure a GEM port:

```
ZXAN(config-if)#gemport 1 name gemport1 tcont 1
ZXAN(config-if)#exit
```

3. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 100 tag
ZXAN(config-if)#exit
```

4. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 100 vlan 100
ZXAN(config-if)#exit
```

5. The following example shows how to configure a service channel in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1
ZXAN(gpon-onu-mng 1/5/1:1)#service HSI gemport 1 cos 0 vlan 100
```

6. The following example shows how to configure a user port VLAN:

```
ZXAN(gpon-onu-mng 1/5/1:1)#vlan port eth_0/1 mode tag vlan 100 pri 0
ZXAN(gpon-onu-mng 1/5/1:1)#end
```

7. Save the configurations.

```
ZXAN#write
```

5.3.11 Configuring the GPON Multicast Service

After you configure the GPON multicast service, subscribers can receive multicast service streams.

Prerequisite

- The GPON ONU is authenticated.
- The T-CONT broadband profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface gpon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **tcont** command to configure the T-CONT.
3. In **ZXAN(config-if)#** mode, run the **gemport** command to configure a GEM port.
4. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.

6. In **ZXAN(config-if)#** mode, run the **service-port** command to configure a service port VLAN.
7. In **ZXAN(config)#** mode, run the **igmp enable** command to enable the IGMP protocol globally.
8. In **ZXAN(config-if)#** mode, run the **igmp fast-leave** command to configure the port IGMP parameters for the ONU.
9. In **ZXAN(config-if)#** mode, run the **igmp mvlan** command to configure an MVLAN.
10. In **ZXAN(config-if)#** mode, run the **igmp mvlan group** command to configure an MVLAN multicast group.
11. In **ZXAN(config-if)#** mode, run the **igmp mvlan source-port** command to configure an MVLAN source port.
12. In **ZXAN(config-if)#** mode, run the **igmp mvlan receive-port** command to configure an MVLAN receiving port.
13. In **ZXAN(config)#** mode, run the **pon-onu-mng gpon-onu** command to enter ONU remote management mode.
14. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **service** command to configure a service channel.
15. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **mvlan** command to configure ONU user port MVLAN.
16. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **vlan port** command to configure an ONU user port VLAN.

– End of Steps –

Example

1. The following example shows how to configure the T-CONT in ONU interface mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 2 name T2 profile 5M
```
2. The following example shows how to configure a GEM port:

```
ZXAN(config-if)#gemport 2 name gemport2 tcont 2
ZXAN(config-if)#exit
```
3. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 200 tag
ZXAN(config-if)#exit
```
4. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 2 vport 2 user-vlan 200 vlan 200
```

```
ZXAN(config-if)#exit
```

5. (Optional) The following example shows how to enable the IGMP protocol globally:

```
ZXAN(config)#igmp enable
```

6. The following example shows how to configure the port IGMP parameters:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#igmp fast-leave enable vport 2
ZXAN(config-if)#exit
```

7. The following example shows how to configure an MVLAN:

```
ZXAN(config)#igmp mvlan 200
```

8. (Optional) The following example shows how to configure the MVLAN operating mode:

```
ZXAN(config)#igmp mvlan 200 work-mode proxy
```

9. The following example shows how to configure an MVLAN multicast group:

```
ZXAN(config)#igmp mvlan 200 group 224.1.1.1 to 224.1.1.3
```

10. The following example shows how to configure an MVLAN source port:

```
ZXAN(config)#igmp mvlan 200 source-port gei_1/20/1
```

11. The following example shows how to configure an MVLAN receiving port:

```
ZXAN(config)#igmp mvlan 200 receive-port gpon-onu_1/5/1:1 vport 2
```

12. The following example shows how to configure a service channel in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1
ZXAN(gpon-onu-mng 1/5/1:1)#service multicast gempport 2 cos 5 vlan 200
```

13. The following example shows how to configure a user port MVLAN:

```
ZXAN(gpon-onu-mng 1/5/1:1)#mvlan 200
ZXAN(gpon-onu-mng 1/5/1:1)#mvlan tag eth_0/2 strip
```

14. The following example shows how to configure a user port VLAN:

```
ZXAN(gpon-onu-mng 1/5/1:1)#vlan port eth_0/2 mode tag vlan 200 pri 5
ZXAN(gpon-onu-mng 1/5/1:1)#end
```

15. Save the configurations.

```
ZXAN#write
```

5.3.12 Configuring the GPON Voice Service (SIP)

After you configure the [GPON](#) voice service, subscribers can make and answer phone calls.

Prerequisite

- The GPON [ONU](#) is authenticated.
- The T-CONT broadband profile is configured.
- The GPON [VoIP](#) IP profile is configured.

- The GPON VoIP VLAN profile is configured.
- The GPON VoIP SIP profile is configured.

Steps

1. In **ZXAN(config)#** mode, run the **interface gpon-onu** command to enter ONU interface configuration mode.
2. In **ZXAN(config-if)#** mode, run the **tcont** command to configure the T-CONT.
3. In **ZXAN(config-if)#** mode, run the **gemport** command to configure a GEM port.
4. In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
5. In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.
6. In **ZXAN(config-if)#** mode, run the **service-port** command to configure a service port VLAN.
7. In **ZXAN(config)#** mode, run the **pon-onu-mng gpon-onu** command to enter ONU remote management mode.
8. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **service** command to configure a service channel.
9. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **voip protocol** command to configure the VoIP protocol type.
10. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **voip-ip** command to configure the VoIP address.
11. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **sip-service** command to configure the VoIP services.

– End of Steps –

Example

1. The following example shows how to configure the T-CONT in ONU interface mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 3 name voip profile 2M
```

2. The following example shows how to configure a GEM port:

```
ZXAN(config-if)#gemport 3 name gemport3 tcont 3
ZXAN(config-if)#exit
```

3. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 300 tag
ZXAN(config-if)#exit
```

4. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 3 vport 3 user-vlan 300 vlan 300
ZXAN(config-if)#exit
```

- The following example shows how to configure a service channel in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1
ZXAN(gpon-onu-mng 1/5/1:1)#service voip-sip gemport 3 cos 7 vlan 300
```

- (Optional) The following example shows how to configure the VoIP protocol type:

```
ZXAN(gpon-onu-mng 1/5/1:1)#voip protocol sip
```

- The following example shows how to the VoIP address:

```
ZXAN(gpon-onu-mng 1/5/1:1)#voip-ip mode static ip-profile ip-test ip-address 1.2.3.4
mask 255.255.255.0 vlan-profile vlan-test
```

- The following example shows how to configure the VoIP services:

```
ZXAN(gpon-onu-mng 1/5/1:1)#sip-service pots_0/1 profile sip-test userid 12345
username 12345 password 12345
ZXAN(gpon-onu-mng 1/5/1:1)#end
```

- Save the configurations.

```
ZXAN#write
```

5.3.13 Configuring the GPON Voice Service (H.248)

After you configure the [GPON](#) voice service, subscribers can make and answer phone calls.

Prerequisite

- The GPON [ONU](#) is authenticated.
- The T-CONT broadband profile is configured.
- The GPON VoIP IP profile is configured.
- The GPON VoIP VLAN profile is configured.
- The GPON VoIP [MGC](#) profile is configured.

Steps

- In **ZXAN(config)#** mode, run the **interface gpon-onu** command to enter ONU interface configuration mode.
- In **ZXAN(config-if)#** mode, run the **tcont** command to configure the T-CONT.
- In **ZXAN(config-if)#** mode, run the **gemport** command to configure a GEM port.
- In **ZXAN(config)#** mode, run the **interface gei** command to enter uplink interface configuration mode.
- In **ZXAN(config-if)#** mode, run the **switchport vlan** command to configure a VLAN.

6. In **ZXAN(config-if)#** mode, run the **service-port** command to configure a service port VLAN.
7. In **ZXAN(config)#** mode, run the **pon-onu-mng gpon-onu** command to enter ONU remote management mode.
8. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **service** command to configure a service channel.
9. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **voip protocol** command to configure the VoIP protocol type.
10. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **voip-ip** command to configure the VoIP address.
11. In **ZXAN(gpon-onu-mng 1/x/x:x)#** mode, run the **mge-service** command to configure the VoIP services.

– End of Steps –

Example

1. The following example shows how to configure the T-CONT in ONU interface mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 3 name voip profile 2M
```

2. The following example shows how to configure a GEM port:

```
ZXAN(config-if)#gemport 3 name gemport3 tcont 3
ZXAN(config-if)#exit
```

3. The following example shows how to configure an uplink port VLAN in uplink interface configuration mode:

```
ZXAN(config)#interface gei_1/20/1
ZXAN(config-if)#switchport vlan 300 tag
ZXAN(config-if)#exit
```

4. The following example shows how to configure a service port VLAN in ONU interface configuration mode:

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 3 vport 3 user-vlan 300 vlan 300
ZXAN(config-if)#exit
```

5. The following example shows how to configure a service channel in remote ONU management mode:

```
ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1
ZXAN(gpon-onu-mng 1/5/1:1)#service voip-h248 gemport 3 cos 7 vlan 300
```

6. The following example shows how to configure the VoIP protocol type:

```
ZXAN(gpon-onu-mng 1/5/1:1)#voip protocol h248 domain iad.zte.com.cn
```

7. The following example shows how to configure the VoIP address:

```
ZXAN(gpon-onu-mng 1/5/1:1)#voip-ip mode static ip-profile ip-test ip-address 1.2.3.4
```

```
mask 255.255.255.0 vlan-profile vlan-test
```

8. The following example shows how to configure the VoIP services:

```
ZXAN(gpon-onu-mng 1/5/1:1)#mgc-service pots_0/1 profile mgc-test  
ZXAN(gpon-onu-mng 1/5/1:1)#end
```

9. Save the configurations.

```
ZXAN#write
```

5.4 Configuring the NTP Service

The ZXA10 C300/C350/C320 is operating in NTP client mode to maintain synchronized time with the NTP server.

Steps

1. In **ZXAN(config)#** mode, run the **ntp enable** command to enable the NTP function.
2. In **ZXAN(config)#** mode, run the **ntp server** command to configure the NTP server.
3. In **ZXAN(config)#** mode, run the **ntp source** command to configure the source address.
4. In **ZXAN(config)#** mode, run the **ntp alarm-threshold** command to configure the alarm threshold.
5. In **ZXAN(config)#** mode, run the **ntp poll-interval** command to configure the interval for synchronizing time with the NTP server.

– End of Steps –

Example

1. The following example shows how to enable the NTP function in global configuration mode:

```
ZXAN(config)#ntp enable
```

2. The following example shows how to configure the NTP server:

```
ZXAN(config)#ntp server 1.2.1.1 priority 2
```

3. The following example shows how to configure the source address contained in the time synchronization request sent by the ZXA10 C300/C350/C320:

```
ZXAN(config)#ntp source vlan1000
```



Note:

The source IP address that the ZXA10 C300/C350/C320 uses to send NTP packets is the inband NM IP address of the ZXA10 C300/C350/C320.

4. The following example shows how to configure the alarm threshold for the NTP synchronization time offset:

```
ZXAN(config)#ntp alarm-threshold 10
```

5. The following example shows how to configure the interval for NTP time synchronization:

```
ZXAN(config)#ntp poll-interval 10
```

6. (Optional) The following example shows how to display the NTP operating status:

```
ZXAN(config)#show ntp status
```

```
Clock is unsynchronized , stratum 5, reference is LOCAL  
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**16  
reference time is 15:30:12 Tue Nov 22 2016 UTC(dbdee304.0)  
clock offset is 0.00, root delay is 0.00  
root dispersion is 6.42, peer dispersion is 5.38
```

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Glossary

CLI

- Command Line Interface

CoS

- Class of Service

CVLAN

- Customer Virtual Local Area Network

DHCP

- Dynamic Host Configuration Protocol

EPON

- Ethernet Passive Optical Network

GEM

- GPON Encapsulation Method

GPON

- Gigabit Passive Optical Network

LLID

- Logical Link Identifier

MGC

- Media Gateway Controller

NE

- Network Element

NM

- Network Management

NMS

- Network Management Server

NTP

- Network Time Protocol

OLT

- Optical Line Terminal

ONU

- Optical Network Unit

P2P

- Peer-to-Peer

PON

- Passive Optical Network

PPPoE

- Point to Point Protocol over Ethernet

RTP

- Real-time Transport Protocol

SIP

- Session Initiation Protocol

SNMP

- Simple Network Management Protocol

SSH

- Secure Shell

SVLAN

- Service Virtual Local Area Network

TID

- Terminal Identification

TLS

- Transparent LAN Service

VLAN

- Virtual Local Area Network

VoIP

- Voice over Internet Protocol

XG-PON

- 10-Gigabit-Capable Passive Optical Network